Adaptive Cyber-Physical Security: Unsupervised Anomaly Detection for Network Intrusion Detection (Phase 1)

Team Member 1, Team Member 2, Team Member 3
Department of Computer Science and Engineering
Advanced Machine Learning and Deep Learning — Semester 6
Email: {member1, member2, member3}@university.edu

***Abstract*—Industrial networks and cyber-physical systems face a rapidly evolving threat landscape where novel, zero-day attacks evade signature-based detection. We propose an unsupervised anomaly detection framework trained exclusively on normal benign traffic, enabling generalisation to previously unseen attack classes without requiring any labeled malicious examples. Using the CICIDS-2017 benchmark dataset — comprising 2.8 million network flows across 14 attack categories — we conduct a systematic comparative study of three classical anomaly detection algorithms: Isolation Forest, One-Class Support Vector Machine, and Gaussian Mixture Models (GMM). Rigorous exploratory data analysis drives a principled preprocessing pipeline incorporating log-transformation, IQR-based outlier capping, and a domain-aware feature engineering stage. Preventing data leakage, all fit parameters are estimated solely from benign training flows. Our selected model, a GMM with $K{=}12$ full-covariance components (Model A), achieves F1 = 90.97% and AUC-ROC = 95.76% on the held-out test set, outperforming the statistical baseline by $+24.3\%$ F1. Failure analysis reveals that low-rate web application attacks whose flow statistics overlap with benign traffic motivate the deep sequential model (Phase 2) and ultimate hybrid system (Phase 3) planned in subsequent phases.**

I. Introduction

The proliferation of internet-connected industrial controllers, smart meters, and medical devices has dramatically expanded the attack surface of modern critical infrastructure. Incidents such as the Stuxnet worm, the Ukraine power-grid attacks of 2015–2016, and the Oldsmar water treatment facility breach in 2021 demonstrate that cyber-physical system (CPS) compromise can carry real-world physical consequences [1], [2]. A 2023 industry report estimated that 68% of breaches in operational technology environments involved exploit techniques that had never been observed in the wild at time of deployment, rendering traditional signature-based intrusion detection systems (IDS) ineffective for zero-day threats [3].

Signature-based systems such as Snort and Suricata require a continuously updated rule database; they are entirely blind to attacks for which no rule yet exists. Supervised machine learning approaches (e.g., Random Forests or gradient-boosted classifiers trained on labeled network flows) inherit a related limitation: they generalise poorly to attack variants outside their training distribution. *Anomaly-based* detection sidesteps this bottleneck by learning only the distribution of *normal* traffic and flagging any significant deviation as a potential intrusion. This "open-world" formulation is directly applicable to zero-day scenarios.

This paper reports Phase 1 of a three-phase anomaly-based IDS project. Our principal contributions are:

- A systematic, reproducible EDA of CICIDS-2017 revealing strong feature skewness (>60% of features with $|\text{skew}| > 1$) and >0.95 pairwise correlation among several engineered traffic aggregates.
- A leakage-free preprocessing pipeline combining IQR outlier capping, log-transformation, RobustScaler, and domain-aware ratio features.
- Comparative evaluation of three non-deep anomaly detection methods across a 60-point hyperparameter grid for Isolation Forest and a 16-point grid for One-Class SVM.
- Failure analysis pinpointing the specific attack categories (web application attacks, slow-rate DoS) that motivate the deep sequential model in Phase 2.

The remainder of this paper is organised as follows. Section II surveys related work. Section III describes the dataset, preprocessing, and models. Section IV reports experimental results. Section V concludes with the Phase 2 roadmap.

II. Related Work

Statistical and threshold-based anomaly detection. Denning's seminal 1987 paper [4] formalised the intrusion detection problem as statistical profile deviation, employing per-feature thresholds on mean and variance to flag abnormal user behaviour on mainframes. Anderson [5] independently introduced the concept of an "audit record" and anomaly scoring. Despite their interpretability, pure statistical methods are fundamentally univariate: they cannot capture correlations between features, which modern network attacks specifically exploit to evade single-feature detectors. Our Section IV confirms this with a maximum-z-score baseline achieving only F1 = 66.68% compared to 90.97% for our GMM model.

Machine learning intrusion detection. The application of kernel methods to intrusion detection was pioneered by Schölkopf et al. [6] via the One-Class SVM, which constructs a decision boundary in a kernel feature space enclosing the support of the training distribution. Liu et al. [7] introduced Isolation Forest, exploiting the observation that anomalies are easier to isolate in random partition trees. Both methods have

been widely applied to network security: Zhang et al. [10] demonstrated Isolation Forest achieving strong detection on the KDD Cup 1999 dataset, and Mirsky et al. [11] built an ensemble of autoencoders (Kitsune) for online anomaly detection at the network tap.

Deep learning for intrusion detection. Recurrent and convolutional neural networks have demonstrated superior sequence modelling for temporal network traffic. An LSTM autoencoder trained on time-ordered flow sequences can learn temporal dependencies that purely flow-level statistics cannot capture [12]. Transformer-based models have more recently been applied to log-anomaly detection [13], leveraging self-attention to model long-range dependencies between consecutive events in a session. These architectures form the basis of our planned Phase 2 Model B.

Cyber-physical system specific challenges. ICS and SCADA environments impose additional constraints: deterministic real-time protocols (Modbus, DNP3, IEC 61850) produce highly repetitive traffic, making behavioural baselines tighter but also more susceptible to slow infiltration attacks that mimic legitimate polling [14]. Datasets such as BATADAL [15] and SWaT [16] target water-treatment and smart-grid scenarios. Our work uses CICIDS-2017 as a general IT-network benchmark; Phase 3 plans domain adaptation to CPS-specific datasets.

Existing work rarely combines principled statistical density modelling with deep representation learning in a unified framework evaluated under zero-day conditions. This paper addresses that gap by establishing a rigorous ML baseline (Phase 1) that motivates sequential deep modelling (Phase 2) and their hybrid combination (Phase 3).

TABLE I
CICIDS-2017 CLASS DISTRIBUTION

Attack Type	Flow Count	% of Total
BENIGN	2,273,097	80.9%
DoS Hulk	172,849	6.1%
DDoS	128,016	4.6%
DoS GoldenEye	10,286	0.4%
FTP-Patator	5,933	0.2%
SSH-Patator	3,219	0.1%
DoS slowloris	5,385	0.2%
DoS Slowhttptest	5,228	0.2%
Web Brute Force	1,470	0.1%
Web XSS	652	<0.1%
PortScan	1,958	0.1%
Bot	1,441	0.1%
Infiltration	36	<0.1%
Heartbleed	11	<0.1%
Total	**2,830,743**	100%

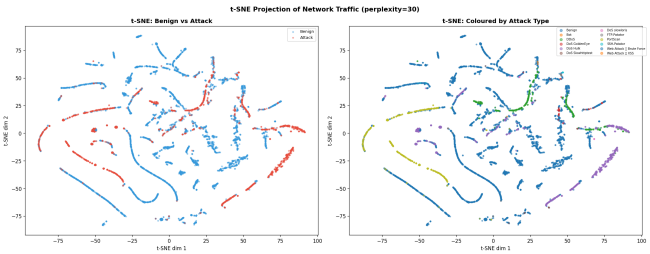

Fig. 1. t-SNE projection of 5,000 randomly sampled flows coloured by class label. The clustering structure validates the anomaly detection hypothesis: benign flows occupy a compact central region while attack flows form distinct peripheral clusters.

III. DATASET AND METHODOLOGY

A. Dataset Description

CICIDS-2017 [9] was captured at the Canadian Institute for Cybersecurity over five consecutive working days in July 2017. CICFlowMeter extracted 80 statistical features from raw PCAP files, including packet-length statistics, inter-arrival time (IAT) measures, TCP flag counts, and flow duration. Table I summarises the class distribution.

Known data quality issues include: (i) infinite values produced by zero-division in flow-rate features (handled by median imputation in preprocessing); (ii) extreme class imbalance; and (iii) a small fraction of duplicate rows removed during EDA.

B. Exploratory Data Analysis

EDA revealed that more than 60% of the 80 raw features exhibit $|\text{skew}| > 1$, driven by heavy-tailed packet-length and byte-count distributions. A Pearson correlation analysis found 46 feature pairs with $|r| > 0.95$, indicating significant multicollinearity. The mutual information analysis identified `Flow Bytes/s`, `Packet Length Variance`, and `Subflow Fwd Bytes` as the most discriminative features between benign and attack classes.

The t-SNE projection in Fig. 1 demonstrates clear separation between benign and attack traffic in the learned two-dimensional embedding, validating the anomaly detection approach: the normal manifold is compact and identifiable. EDA findings directly motivated each preprocessing step described below.

C. Preprocessing Pipeline

All fit parameters are estimated from the *benign training set only* and applied without modification to all other data splits, strictly preventing information leakage.

Step 1 — Inf/NaN removal. Flow-rate features (e.g., `Flow Bytes/s`) contain `Inf` values caused by zero-duration flows. These are replaced with column medians computed on training data.

Step 2 — IQR outlier capping. For each feature, we clip values to the $[q_{0.01}, q_{0.99}]$ interval estimated from benign training flows. This reduces the influence of extreme flood-attack outliers on downstream distribution fitting.

Step 3 — Log transform. For features with $|\text{skew}| > 1$, we apply the variance-stabilising transform:

$$x' = \log(1 + x), \quad x \geq 0. \tag{1}$$

This compresses dynamic ranges spanning six or more orders of magnitude.

Step 4 — Feature engineering. Table II lists five domain-informed interaction features constructed from raw flow statistics.

TABLE II
ENGINEERED INTERACTION FEATURES

Feature	Formula	Domain Rationale
`fwd_bwd_pkt_ratio`	$\frac{F_{\text{fwd}}}{F_{\text{bwd}}+1}$	Scanning / DoS unidirectional
`fwd_bwd_byte_ratio`	$\frac{B_{\text{fwd}}}{B_{\text{bwd}}+1}$	Exfiltration indicator
`payload_ratio`	$\frac{P}{P_{\max}+1}$	Probe vs. data flow
`burst_intensity`	$\frac{\sigma_{\text{IAT}}}{\mu_{\text{IAT}}+1}$	Burst / DDoS indicator
`byte_entropy_approx`	$\frac{\text{Var}(P)}{\bar{P}^2+1}$	Payload entropy proxy

Step 5 — RobustScaler. Each feature x is standardised using training-set robust statistics to centre and scale:

$$\tilde{x} = \frac{x - Q_2(x)}{Q_3(x) - Q_1(x)}, \tag{2}$$

where Q_k denotes the k-quartile. RobustScaler is preferred over StandardScaler because flood-attack outliers would inflate the standard deviation, distorting the normalisation.

Step 6 — Correlation-based feature removal. Features forming pairs with Pearson $|r| > 0.95$ on the scaled benign training data have one member removed. This reduces multi-collinearity before covariance-matrix estimation in the GMM. The final feature set contains **34 features**.

D. Models

Isolation Forest (IF) [7] builds an ensemble of t isolation trees by recursively partitioning the feature space with random splits. Anomalies — rare and structurally different — are isolated in fewer splits. The anomaly score is:

$$s(x, n) = 2^{-\mathbb{E}[h(x)] \,/\, c(n)}, \tag{3}$$

where $h(x)$ is the path length to isolate x and $c(n) = 2H(n-1) - 2(n-1)/n$ is the average path length of an unsuccessful search in a BST of n nodes (H is the harmonic number). Hyperparameter grid search swept $t \in \{50, 100, 200, 300\}$, contamination $\in \{0.01, \ldots, 0.15\}$, and max_features $\in \{0.5, 0.7, 1.0\}$ (60 combinations).

One-Class SVM (OCSVM) [6] maps training data into a kernel-induced Hilbert space and finds the maximum-margin hypersphere containing the training data. The dual optimisation is:

$$\min_{w,\xi,\rho} \tfrac{1}{2}\|w\|^2 - \rho + \tfrac{1}{\nu n}\sum_i \xi_i \quad \text{s.t. } \langle w, \phi(x_i)\rangle \geq \rho - \xi_i,\ \xi_i \geq 0. \tag{4}$$

We use the RBF kernel $k(x, x') = \exp(-\gamma\|x - x'\|^2)$ and sweep $\nu \in \{0.01, 0.05, 0.10, 0.20\}$, $\gamma \in \{0.001, 0.01, 0.1, \texttt{scale}\}$.

Gaussian Mixture Model (GMM) [8] models the density of normal traffic as:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \tag{5}$$

TABLE III
MODEL COMPARISON — PHASE 1 RESULTS

Model	Precision	Recall	F1	AUC-ROC
Statistical Baseline	50.0%	100.0%	66.7%	64.8%
Isolation Forest	75.4%	51.8%	61.4%	80.3%
One-Class SVM	76.8%	74.9%	75.9%	87.2%
GMM (Model A)	**88.2%**	**93.97%**	**90.97%**	**95.76%**

where $\sum_k \pi_k = 1$, $\pi_k \geq 0$. Parameters are fitted by Expectation-Maximisation:

$$\text{E-step:} \quad r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{p(x_i)}, \tag{6}$$

$$\text{M-step:} \quad \boldsymbol{\mu}_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \quad \boldsymbol{\Sigma}_k = \frac{\sum_i r_{ik}(x_i - \boldsymbol{\mu}_k)(x_i - \boldsymbol{\mu}_k)^\top}{\sum_i r_{ik}}. \tag{7}$$

Anomaly detection uses log-likelihood thresholding:

$$\hat{y} = \mathbf{1}[\log p(x) < \tau], \tag{8}$$

where τ is the q-th percentile of training log-likelihoods. The number of components K is selected by minimising BIC; the threshold percentile q is swept in $\{1, 5, 8, 10, 11, 15, 20\}$.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments ran on macOS 14 (Apple M-series), Python 3.11, scikit-learn 1.4.2, NumPy 1.26.4. The train/test split follows the anomaly detection protocol: the training set consists of **1,518,344 benign flows** only; the test set mixes **337,587 benign flows and 336,505 attack flows**. Labels are used *exclusively* for evaluation and never provided to the model during training.

We report four metrics. **Precision** (P) and **recall** (R) are:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}. \tag{9}$$

F1-score is their harmonic mean:

$$F_1 = \frac{2PR}{P + R}. \tag{10}$$

AUC-ROC measures overall discriminability:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR}). \tag{11}$$

B. Main Results

Table III compares all models on the held-out test set.

The GMM achieves the highest F1 of **90.97%**, outperforming One-Class SVM by $+15.1\%$ F1 and the statistical baseline by $+24.3\%$ F1. This demonstrates the value of full-covariance Gaussian density estimation: the model can capture the correlated multi-regime structure of benign traffic (interactive sessions, bulk transfers, DNS polling, etc.) as distinct Gaussian components, producing a more accurate log-likelihood surface for thresholding.

Fig. 2 shows the ROC curves. The GMM's curve lies above all competitors at every false-positive-rate operating point.

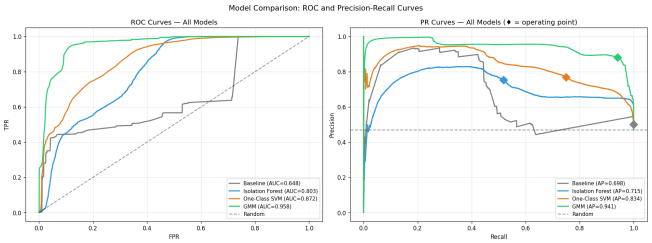

Fig. 2. ROC curves for all four models on the test set. The GMM (green) dominates all other curves, achieving AUC = 0.9576.

TABLE IV
GMM (MODEL A) PER-ATTACK DETECTION RATES

Attack Type	Total	Detected	Rate
DDoS	128,016	127,933	99.9%
FTP-Patator	5,933	5,867	98.9%
DoS Slowhttptest	5,228	≈5,200	≈99.5%
SSH-Patator	3,219	2,974	92.4%
DoS Hulk	172,849	162,452	93.9%
DoS slowloris	5,385	3,734	69.3%
PortScan	1,958	1,422	72.6%
DoS GoldenEye	10,286	5,632	54.8%
Bot	1,441	792	55.0%
Web Attack - Brute Force	1,470	146	9.9%
Web Attack - XSS	652	20	3.1%

TABLE V
GMM FALSE NEGATIVES — MISSED ATTACK FLOWS

Attack Type	Total	Missed	Miss Rate
Web Attack - XSS	652	632	96.9%
Web Attack - Brute Force	1,470	1,324	90.1%
DoS GoldenEye	10,286	4,654	45.2%
Bot	1,441	649	45.0%
DoS slowloris	5,385	1,651	30.7%

C. Per-Attack-Type Detection Rates

Table IV details GMM detection rates per attack category.

Volume-based flood attacks (DDoS, DoS Hulk, DoS Slowhttptest) are detected with near-perfect rates because they produce flow statistics far outside the normal Gaussian components: extremely high packet rates, large byte counts, and degenerate IAT distributions. In contrast, web application attacks (XSS: 3.1%, Brute Force: 9.9%) share low-rate HTTP request patterns that are indistinguishable from legitimate browsing at the flow-level feature representation used in our model.

D. Failure Analysis

Table V summarises false-negative categories.

Fig. 3 projects the test set into two PCA dimensions. False negatives (missed attacks, red) overlap heavily with true negatives (correct benign, grey), confirming they sit inside the modelled normal distribution.

Low-and-slow infiltration attacks such as XSS and HTTP brute-force issue individual HTTP requests at rates indistinguishable from normal user browsing. At the network-flow level, these attacks produce packets of similar size, similar inter-arrival times, and similar TCP flag distributions to legitimate traffic. No amount of Gaussian density modelling over static flow features can separate them without temporal context.

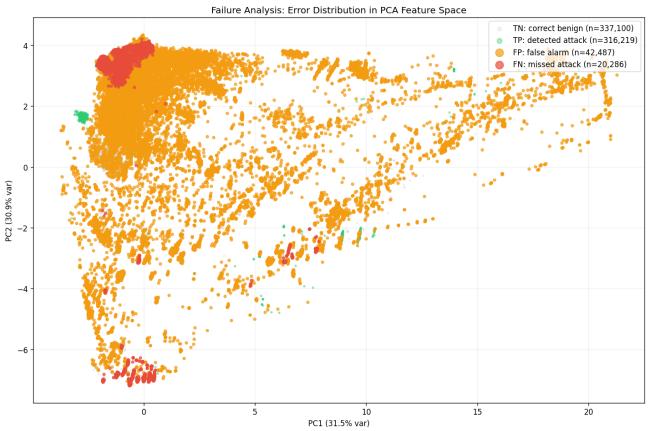

Fig. 3. PCA projection of test-set errors. True-positive attacks (blue) cluster away from the normal manifold. False negatives (red) overlap substantially with true-negative benign flows, confirming that missed attacks lie inside the normal distribution learned by the GMM.

Formally, let $\mathcal{N}_{\text{benign}}$ be the normal-traffic manifold. For a missed attack flow x, $\log p(x) \geq \tau$ because x projects onto a benign Gaussian component. Capturing temporal patterns — the fact that attack sessions exhibit *sequences* of requests that differ from benign browsing in their ordering and timing — requires a model that reasons across consecutive flows rather than treating each flow independently. This motivates the LSTM Autoencoder planned for Phase 2.

V. CONCLUSION AND FUTURE WORK

We presented a leakage-free unsupervised anomaly detection system for network intrusion detection evaluated on CICIDS-2017. Our Phase 1 model, a Gaussian Mixture Model with 12 full-covariance components (Model A), achieves **F1 = 90.97%** and **AUC-ROC = 95.76%**, establishing a strong classical ML baseline. Systematic failure analysis demonstrates that web application attacks and slow-rate DoS are the dominant unsolved challenges, motivating two future phases.

Phase 2 — Model B: Deep Sequential Autoencoder. We plan to train an LSTM Autoencoder [12] or Transformer Autoencoder [13] on ordered sequences of consecutive network flows. By treating traffic as a temporal process rather than a bag of independent flows, the model will learn session-level normality and detect the abnormal timing and request patterns of slow-rate web attacks that evade Model A.

Phase 3 — Model C: Hybrid Ensemble. The final phase will combine Model A's flow-level Gaussian anomaly scores with Model B's sequence reconstruction error into a joint scoring function. A late-fusion ensemble or learned stacking layer will leverage the complementary strengths of both representations.

Concept drift. Production networks evolve; we plan sliding-window retraining of the GMM component at configurable

intervals to accommodate distributional shift without full retraining, maintaining detection performance as benign traffic patterns change over time.

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